

1. In a Buck-Boost converter, consider all components to be ideal. Let input voltage V_d be 12 ~ 48 V, output voltage $V_o = 24$ V, and $f_s = 50$ kHz, $L = 50$ μ H. (17 points)

理想条件下，Buck-Boost 变换器的输入电压 V_d 为 12 ~ 48 V，输出电压 V_o 为 24 V，开关频率为 50 kHz，电感值为 50 μ H。(17 分)

- (1) Draw the Buck-Boost converter circuit diagram. (3 points)

画出 Buck-Boost 变换器的电路图。(3 分)

- (2) Draw the waveforms of v_L 、 i_L 、 i_C when the converter operating in a continuous-conduction mode. (3 points)

当变换器工作在 CCM 模式时，画出电感电压、电感电流 和电容电流 的波形图。(3 分)

- (3) Calculate the voltage stress of switch and diode respectively. (2 points)

分别计算开关管和二极管电压应力。(2 分)

- (4) Calculate the minimum P_o that will keep the converter operating in a continuous-conduction mode for all range of V_d . (7 points)

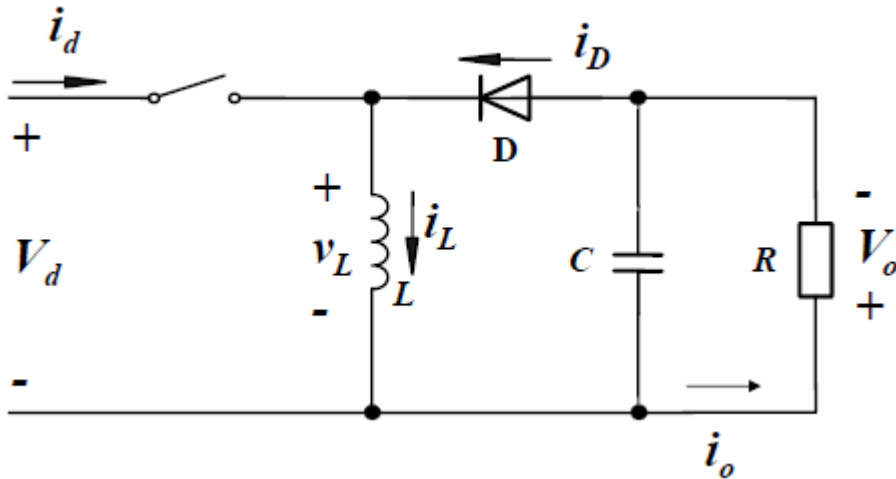
试计算在全输入电压范围内能够变换器保持工作在 CCM 模式的最小输出功率 P_o 。(7 分)

- (5) What will happen if keep $D = 0.5$ and the load R is disconnected? (2 points)

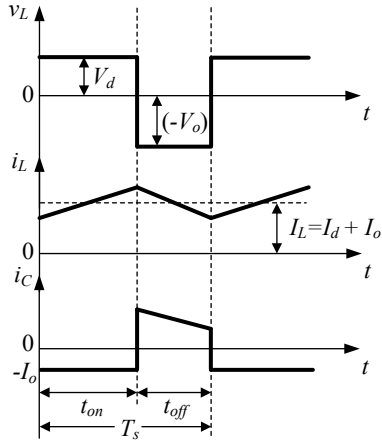
如果在负载电阻 R 断开时仍保持占空比 $D = 0.5$ ，则将会发生什么？(2 分)

答：

- (1) Buck-Boost 变换器电路图 (3 分)：



- (2) v_L 、 i_L 、 i_C 波形 (3 分)：



(3)

Switch 电压应力: $V_o + V_{d\max} = 72 \text{ V}$ (1 分)

Diode 电压应力: $V_o + V_{d\max} = 72 \text{ V}$ (1 分)

(4) 确定占空比 D 范围

$$QD = \frac{V_o}{V_o + V_d} \quad (1 \text{ 分})$$

$$\therefore D_{\min} = \frac{1}{3}, D_{\max} = \frac{2}{3} \quad (1 \text{ 分})$$

$$QI_{LB} = \frac{i_{L,peak}}{2} = \frac{T_s V_d}{2L} D = \frac{T_s V_o}{2L} (1 - D)$$

$$I_{oB} = (1 - D)I_{LB} = \frac{T_s V_o}{2L} (1 - D)^2 \quad (1 \text{ 分})$$

当 $D_{\min} = \frac{1}{3}$ 时, 临界输出电流取最大值, 得 $I_{oB\max} = 2.133 \text{ A}$ (2 分)

如果要工作于 CCM 模式, 则 $I_{o\min} \geq I_{oB\max}$, 因此确保工作于 CCM 模式的输出功率最小值为:

$$P_{o\min} = I_{oB\max} \times V_o = 51.2 \text{ W} \quad (2 \text{ 分})$$

(5)

当负载开路时, 电容上的能量没有释放回路, 且电感不断给电容充电, 使得电容两端的电压不断抬升, 最终击穿电容及其它器件。 (2 分)

2. A Forward converter with a demagnetizing winding ($N_3=2N_1$) operates in a continuous-conduction mode, $V_d=45\sim60\text{V}$, $V_o=30\text{V}$, $f_s=100\text{kHz}$, P_{load} is $30\text{W}\sim60\text{W}$.

Assume all components to be ideal except the transformer magnetization inductance.

Please answer the following questions: (16 points)

一个正激变换器工作在电感电流连续模式，其磁通复位绕组匝数 $N_3=2N_1$, $V_d = 45\sim 60V$, $V_o=30V$, $f_s=100kHz$, P_{load} 为 $30W\sim 60W$ 。除了变压器励磁电感，假设变换器中所有元器件均为理想元件。请回答以下问题。(16 分)

(1) Draw the circuit diagram of this converter.

请画出正激变换器电路图。(4 分)

(2) Please derive the relationship between V_o and V_d .

请推导输出电压 V_o 与输入电压 V_d 之间的关系。(3 分)

(3) Calculate N_2/N_1 if this turns ratio is desired to be as small as possible.

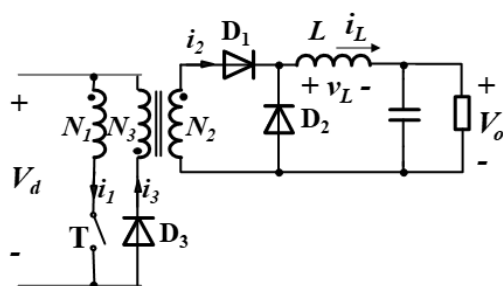
计算 N_2/N_1 的最小值。(4 分)

(4) With the value of N_2/N_1 from (3), calculate the minimum value of the filter inductance L .

采用(3)问中的 N_2/N_1 数值，计算滤波电感 L 的最小值。(5 分)

参考答案:

(1)



(4 分)

(同名端画错扣 2 分；二极管方向不对，扣 2 分；叠加扣分，两个都不对，就是 0 分)

(2)

$$\because \left(\frac{N_2}{N_1} V_d - V_o \right) T_{on} - V_o T_{off} = 0 \quad (3 \text{ 分})$$

$$\therefore \frac{V_o}{V_d} = \frac{N_2}{N_1} D$$

(3)

$$\therefore D_{\max} = \frac{1}{1 + \frac{N_3}{N_1}} = \frac{1}{3}$$

$$\frac{V_o}{V_d} = \frac{N_2}{N_1} D \quad (4 \text{ 分})$$

$$\therefore \frac{N_1}{N_2} = D \frac{V_d}{V_o} \leq \frac{1}{3} \frac{V_{d\min}}{V_o} = \frac{1}{2}$$

$$\therefore \frac{N_2}{N_1} \geq 2$$

(4) 为了保持连续, 需要 $I_L \geq 0.5 \Delta i_L$, 有

$$I_{o\min} = \frac{P_{o\min}}{V_o} = I_{L\min} \geq \frac{1}{2} \frac{\left(V_d \frac{N_2}{N_1} - V_o \right) D T_s}{L}$$

$$L \geq \frac{1}{2} \frac{\left(V_d \frac{N_2}{N_1} - V_o \right) D}{f_s I_{o\min}} = \frac{1}{2} \frac{V_o (1-D)}{f_s I_{o\min}}$$

$$\therefore \frac{N_2}{N_1} = 2 \Rightarrow \frac{V_o}{V_{d\max}} \leq \frac{N_2}{N_1} D \leq \frac{V_o}{V_{d\min}} \Rightarrow \frac{1}{4} \leq D \leq \frac{1}{3}$$

$$\therefore L_{\min} = \frac{1}{2} \frac{V_o (1-D_{\min})}{f_s I_{o\min}} = 112.5 \mu H$$

(给出 L 表达式, 给 3 分; 计算出结果给 2 分)

3. (18 points) A Forward converter is to be designed with the following specifications:

Input voltage range $V_d = 20V \sim 28V$, output voltage $V_o = 100V$, switching frequency $f_s = 100kHz$, working in CCM, N_1 is the turns of primary winding, N_2 is the turns of secondary winding, the demagnetizing winding turns $N_3 = 2N_1$. Assume all components to be ideal except the presence of transformer magnetization inductance.

(1) Please draw the circuit diagram of the Forward converter with demagnetizing winding. (3 points)

(2) Derive the maximum duty ratio $D_{\max}$. (3 points)

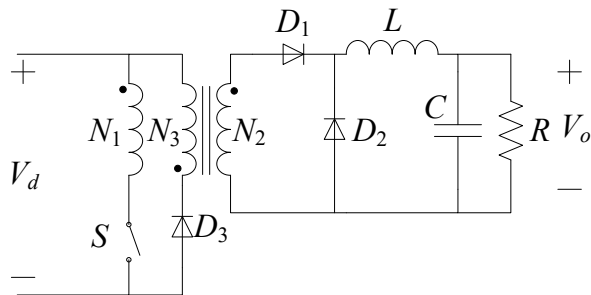
(3) Calculate the minimum turns ratio N_2/N_1 of transformer. (4 points)

(4) Calculate the voltage stress of the switch? (2 points)

(5) Draw the voltage waveform across the output filter inductor v_L and the voltage

waveform of switch v_T when $V_d=20V$, and mark the value. (6 points)

(1) Draw circuit (2 分), Mark dotted terminals(1 分)



(2)

$$D_{\max} = \frac{t_{on}}{t_{on} + t_m} = \frac{1}{1 + \frac{N_3}{N_1}} = \frac{1}{1+2} = \frac{1}{3} \quad (3 \text{ 分})$$

(3)

$$\frac{V_o}{V_d} = \frac{N_2}{N_1} D \quad \frac{N_2}{N_1} = \frac{V_o}{DV_d} = 3 \times 100 / 20 = 15 \quad (4 \text{ 分})$$

(4)

$$V_T = (1 + N_3/N_1) \times V_d = (1+2) \times 28 = 84V \quad (2 \text{ 分})$$

(5) Waveforms: (4 分)

